

Solutions for the energy sector, including renewables, in general, will be most rapidly available as companies embrace the inter-market technology advancement that is occurring today. All companies seeking new technologies need to only look at parallel markets and adopt technologies for their specific needs. For example, sensing technology is now available for most constituents of energy exploration and extraction. It is now being ruggedised to enable application in the harsh environment of energy exploration. Companies are already working on this. For example, Embitel Technologies from India was selected as one among top ten companies to present Internet of Things (IoT) solutions at Oil and Gas CleanTech Challenge in Colorado, USA, under the digital oil field category.

Doctoral student Joseph Carr developed an electrical power-converter system that can simultaneously accept power from multiple energy sources and convert it for use in the electrical grid. "The researchers' high-frequency matrix converter addresses these shortcomings. Its simplified control system uses power converters to allow connection of a variety of power sources to a small, high-frequency transformer. Then, using a high-frequency matrix converter, it produces stable electricity ready to be supplied to the electrical grid system," explains a report in *Science Daily*.

"In data centre server racks, power requirement is based on the server situations. Enabling customers to have access to power demand per server rack allows them to better optimise the systems," explains Simarpreet Singh, head strategy, Hartek Power Pvt Ltd.

National Renewable Energy Laboratory CSP group manager Mark Mehos, co-author with Jennie Jorgenson and Paul Denholm of the study, finds that concentrating solar power (CSP), with its ability to store energy for several hours or more, helps maintain firm capacity in those hours when the Sun is below the horizon.

Can renewables survive against cheaper fossil fuels

Fossil fuel has been the predominant energy source for over a century. This trend will continue for decades to come. A peaceful balance between fossil fuels and renewables will be found when technology enables a clear crossover opportunity that makes all three streams of fossil fuel development clean. The fossil fuel sector is turning clean-tech as technology enables real-time monitoring of critical processes, enabling a cleaner extraction, processing and delivery ecosystem. Integrating key sensor monitoring systems into real-time decision tools through the IoT is the essential ingredient that Embitel is delivering to the market.

If you look at the period of overall rising oil prices, renewable energy saw a spike due to climate change discussions. "India is vulnerable to oil prices because we are dependent on oil for 70 per cent of requirements and that is almost entirely made up of what we import. Thermal and nuclear energy programmes are mostly stuck because of regulatory problems, and we have seen solar and wind to be the driving areas. All this has changed with the shale gas situation, which brought in a scenario where the USA reached a point of energy self-sufficiency, to the point that they can actually look at exports. The Middle-Eastern bloc has an agenda to keep prices low in order to affect viability of shale gas," explains Sanjay Kumar. The result? This brings viability of renewable energy under a question mark. Who will drive it if it is not economically viable?

While photovoltaics convert light into electricity, CSP technologies concentrate the Sun's energy and capture it as heat, which then drives an engine or turbine to produce electrical power. However, NREL explains that thermal energy CSP generates can be held back for several hours via storage systems such as molten salts. These can later be used after the Sun sets.

While SolarCity has the highest efficiency in mass-produced solar cells, the world record is held by a 46-per-cent-efficient multi-junction solar cell developed by Soitec and CEA-Leti, France, together with Fraunhofer Institute for Solar Energy Systems ISE, Germany.

As wind turbines increase in number, atmospheric science professors from University of Kansas, USA, found in August 2015 that wind turbines in large wind farms trigger a slowdown effect. This causes a decrease in the power generated per wind turbine as each additional turbine is installed.

While controlling the number of turbines is one way to prevent this problem, one Spanish start-up may have found a solution. Called Vortex Bladeless, the company is working on building blade-less turbines. It converts oscillations caused by vortices, created by air movement around the structure, into power using

an alternator at the base of the device. The blade-less design reduces manufacturing costs, allows to have a denser installation in a given area and is much more silent than existing wind turbines.

In addition to hardware, there are innovations in the form of control systems that improve renewable energy sources. For instance, researchers at Applied Electronics Research Team have developed an adaptive algorithm that orders the wind turbine to adapt to change in wind speed such that the efficiency is at the maximum.

Connecting renewable energy with communities

Something similar to connected small energy sources can be read about in the sequel to the Social Science fiction book *Daemon*, released in 2006, by author Daniel Suarez. The setting described in the book has self-sufficient communities using natural energy sources and connected technologies to avoid the centralised agency-controlled systems seen today.

What we are working on now is to make the system smart. Everybody has a combination of biomass, wind, solar and biogas. "With the new model, everyone could be a consumer and everyone could be a generator of energy. So any building with surface area available on top